Dan Ai

University of Massachusetts Amherst
Department of Resource Economics
80 Campus Center Way
Amherst, MA 01003 USA

Email: dai@umass.edu
Website: https://danai-econ.com
Phone: +1 (413) 425-3309

Education

University of Massachusetts Amherst — Amherst, USA
Ph.D. Candidate in Resource Economics

Macau University of Science and Technology — Macau, China
M.S. in Applied Economics

Langfang Normal University — Langfang, China
B.S. in Computer Science and Technology

Fields

Energy and Environmental Economics, Applied Microeconomics, Applied Econometrics

Publications

Ai, D., Christine L. Crago, and Jamie T. Mullins (2026). "Heat, Power Outages, and Mortality in the United States." *Journal of Environmental Economics and Management*, 137, 103275.

Hsiao, C.Y., D. Ai, X. Wei, and N. Sheng (2021). "The Contagious Effect of China's Energy Policy on Stock Markets: The Case of Solar PV Industry." *Renewable Energy*, 164, 74–86.

Working Papers

Data Centers and Transmission Constraints
Job Market Paper

Disparities in Power Outage Experience in the United States
With Christine L. Crago, Zeal Shah, and Jay Taneja
(Best Student Conference Paper, Northeastern Agricultural and Resource Economics Association (NAREA), 2024)

Work in Progress

Do Data Centers Warm Their Neighborhoods? Evidence from Satellite Land Surface Temperature

Teaching Experience

Course Instructor, University of Massachusetts Amherst
Price Theory

Teaching Assistant, University of Massachusetts Amherst
Cost Benefit Analysis
Financial Analysis for Consumers and Firms
Managerial Economics
Intermediate Statistics for Business and Economics
Introductory Statistics for the Social Sciences

Teaching Assistant, Macau University of Science and Technology
Microeconomics
Macroeconomics

Presentations

2025: Northeastern Agricultural and Resource Economics Association Annual Meeting; AERE Summer Conference (Santa Ana Pueblo, NM); UMass Energy Transition Symposium (Amherst, MA)

2024: Northeastern Agricultural and Resource Economics Association Annual Meeting (Rehoboth, DE); Sloan/Haas Summer School in Environmental and Energy Economics (University of California, Berkeley, CA)

2023: UMass Energy Transition Symposium (Amherst, MA)

Awards and Fellowships

Graduate Fellows Travel Grant, UMass Amherst Graduate School 2025
Best Student Conference Paper, Northeastern Agricultural and Resource Economics Association (NAREA) 2024
Graduate Fellows Travel Grant, UMass Amherst Graduate School 2024
Research Enhancement and Leadership (REAL) Fellow, UMass Amherst Graduate School 2021–2025

Affiliations

Elevating Equity Values in the Transition of the Energy System (ELEVATE) Program, funded by the U.S. National Science Foundation (NSF)

Skills

Programming: Java, Python, R, Stata, ArcGIS Pro, LaTeX, Google Earth Engine

Miscellaneous

Language: English (Fluent), Chinese (Native)
Citizenship: Chinese

References

Christine L. Crago
Associate Professor
Department of Resource Economics
University of Massachusetts Amherst
ccrago@resecon.umass.edu

Jamie T. Mullins
Associate Professor
Department of Resource Economics
University of Massachusetts Amherst
jmullins@umass.edu

Matt Woerman
Assistant Professor
Department of Agricultural and Resource Economics
Colorado State University
matt.woerman@colostate.edu